

ronment. Even with a modular design, there is a ridged structure supporting the environment that involves more than a step ladder and some pre-terminated cable to adjust. Things like over-head ladder rack or under-floor cable troughs are disruptive to install and change.

No design will support every eventuality, but try to cover the ones that are most likely to occur in your environment. A properly designed ladder rack system allows most changes with little or no rework required and no downtime to make modifications.

Future of data cabling. As computer systems evolve with processors becoming faster, more functionality is being supported on smaller platforms. Virtualisation of everything, including servers, networks, storage and desktops, is pushing flexibility at the infrastructure level. This flexibility requires large quantities of LAN and SAN bandwidth to allow for the peak demands that can be imposed by a virtualised computing world.

The rapid adoption of 10GbE and 8GbE fibre channel, with the promise of 40 and 100 GbE in the next couple of years, questions the viability of Category 5/6 cabling as a method for connecting servers in the future. The size of Category 6 enhanced cabling, along with its installation criteria, needs to be considered both from the financial investment and the physical implementation perspective.

A rack full of blade server chassis will most likely be running over multiple 10Gb connections back to a network switch that may or may not be a hybrid LAN switch. What we do know is that the distance limitation of 10 GbE over copper will limit its use within a rack or between several racks in a row. As fibre continues to expand in usage, the manufacturing price will continue to drop. The price of copper, along with the weight and size of the cables and their distance limitations, will become more of a factor in the data centre. You can expect to provide fibre to every rack in the data centre in the time to come, so planning for it is a requirement.

The new modular fibre systems offer a perfect solution for data centres that are subject to change. By using some of the modular fibre systems available in the market today, you can install new fibre trunks in various densities quickly and with minimum tools. No special fibre-terminating skills or tools are required, as long as the data centre has been designed to support this type of solution.

Power cabling. The point is that where there used to be a pair of circuit breaker panels, you can expect to triple the quantity unless other methods of power distribution are adopted. There are several alternatives to the traditional centralised circuit breaker panel solution. The first, and widely adopted solution for co-location centres, is a zone power distribution system where

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power distribution is provided by zones sometimes as small as a row or pair of rows of servers. The power distribution units are located in close proximity to the racks to which they are supplying power. Their form factor is designed to fit with a row of racks and complement the airflow requirements of a hot-aisle/cold-aisle design.

Another option is a distributed bus system. A bus system is typically run overhead and supports up to 400 amps or more of 3-phase power. Fig. 3 shows a distributed bus system that facilitates various power requirements, improves flexibility and reduces the large quantity of runs from each power distribution unit to each rack.

Distributed bus systems are avail-

able in capacities ranging from 60A, 110V/220V to 400A, 3-phase. The larger the capacity, the more versatile the system. Running these systems overhead leaves more room under the floor for air-flow. Connectivity to the bus is done with special tap devices that can be configured with a variety of outlets to support the plug types required in each rack. For full redundancy, two bus systems can be run for each zone to be supported.

The taps are configured with circuit breakers and monitoring systems to keep track of current and voltage. The feed to the bus system also has electronic monitoring to monitor phase loading and overall capacity.

The advantage to the system is that it allows easy reconfiguration of power provisioning to racks in any designated area of the data centre without having to install more branch circuits and conduit under the raised floor or over-head bolted to the ceiling or ladder rack system. If a rack needs to be relocated and requires special power outlets and provisioning, it is a simple matter of disconnecting the taps and moving them with the rack.

There are significant savings in reducing the amount of cabling and conduit running from the power distribution unit or the circuit breaker panel out across the floor to the racks. In a traditional installation, each three-phase circuit requires four or five conductors—three for the power feeds, one for ground and one for neutral. This means that ten copper wires (five for each circuit) must run from the panels to each rack. In addition, a power trough or individual conduit must also be installed. In some cases, long WIPs are installed, allowing some flexibility in the outlet placement under the floor and reducing the amount of ridged conduit required, but the copper cable runs are still required. ●

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